

English

PENG

October 6, 2026

Contents

1	Vocabulary	1
1.1	adjective	1
1.2	euphoric	1
1.3	epiphany	1
1.4	execute	2
1.5	mutation	2
1.6	imperative	2
1.7	immense	2
1.8	facetious	2
1.9	mantra	2
1.10	auxiliary	2
1.11	permutation	2
1.12	bureaucratic	3
1.13	conundrum	3
1.14	polymorphism	3
1.15	customary	3
1.16	reconcile	3
1.17	tedious	3
1.18	succinct	3

1 Vocabulary

<https://dictionary.cambridge.org/>

1.1 adjective

- noun [ˈædʒ.ek.tɪv]
 - a word that describes a noun or pronoun

1.2 euphoric

- adjective [juːˈfɔːr.ɪk]
 - extremely happy and excited
 - * a euphoric mood

1.3 epiphany

- noun [ˈɪpɪf.ə.n.i]
 - a moment when you suddenly feel that you understand, or suddenly become conscious of, something that is very important to you
 - * Euphoric, mind-bending epiphanies may result!

1.4 execute

- verb [ˈɛk.sɪ.kjuːt]
 - to do or perform something, especially in a planned way
 - * execute instructions

1.5 mutation

- noun [mjuːˈteɪ.ʃən]
 - the way in which genes change and produce permanent differences
 - * No mutation! Everything (variables, data structure) is immutable.

1.6 imperative

- adjective [ɪmˈper.ə.tɪv]
 - extremely important or urgent
 - used for giving an instruction or order
 - * an imperative or object-oriented paradigm

1.7 immense

- adjective [ɪˈmens]
 - extremely large in size or degree
 - * immense help

1.8 facetious

- adjective [fəˈsiː.ʃəs]
 - not serious about a serious subject, in an attempt to be funny or to appear clever
 - * “If it compiles, it must be correct” is mostly facetious.

1.9 mantra

- noun [ˈmæn.trə]
 - a word or phrase that is often repeated and expresses a particular strong believe
 - * “Don’t Repeat Yourself” is a mantra often heard in the world of programming.

1.10 auxiliary

- adjective [ɔːgˈzɪl.i.ə.ri]
 - giving help or support, especially to a more important person or thing
 - * auxiliary boundary conditions [1]

1.11 permutation

- noun [ˌpɜː.mjuːˈteɪ.ʃən]
 - any of the various ways in which a set of things can be ordered
 - * This sorting function returns a permutation of its input.

1.12 bureaucratic

- adjective [ˌbjʊə.rəˈkræt.ɪk]
 - involving complicated rules and processes that make something slow and difficult
 - * The program become long and bureaucratic.

1.13 conundrum

- noun [kəˈnʌn.drəm]
 - a question that is a trick, often involving a humorous use of words that have two meanings
 - * `Nat.zero` is a bit of a conundrum.

1.14 polymorphism

- noun [ˌpɒl.iˈmɔːfɪ.zəm]
 - the fact that something such as an animal or organism can exist in different forms
 - * The term polymorphism typically refers to datatypes and definitions that take types as arguments.

1.15 customary

- adjective [ˈkʌs.tə.mər.i]
 - traditional
 - * It is customary to use Greek letters to name type arguments in Lean.

1.16 reconcile

- verb [ˈrek.ən.saɪl]
 - to find a way in which two situations or beliefs that are opposed to each other can agree and exist together
 - * Programs are difficult to reconcile with the understanding of computation as the evaluation of mathematical expressions.

1.17 tedious

- adjective [ˈtiː.di.əs]
 - boring
 - * Most proofs are written for people to understand, and leave out many tedious details.

1.18 succinct

- adjective [səkˈsɪŋkt]
 - said in a clear and short way; expressing what needs to be said without unnecessary words
 - * A reference work in which all aspects of Lean are clearly specified, and demonstrated through succinct examples.

References

- [1] Lawrence C Evans. *Partial differential equations*. Vol. 19. American mathematical society, 2022. ISBN: 1470469421.